Measurement of the remanent magnetic quadrupole moment of a macroscopic mass using a torsion balance for gravitational experiment

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Magnetic effects significantly contribute to errors in gravitational experiments. Typically, remanent magnetism is characterized by magnetic dipole interactions. However, this approach neglects higher-order magnetic terms, particularly the remanent magnetic quadrupole moment (RMQM) interaction, which may compromise experimental accuracy. This paper focuses on measuring the RMQM of a macroscopic mass to evaluate its impact. Measurements were performed using a rotating torsion pendulum equipped with custom-designed magnetic coils. The investigation examines two predominantly non-magnetic samples, one of which is an optimized pendulum specifically designed to test the equivalence principle. The results indicate that the RMQM components of interest for the pendulum have a magnitude of $3.7(3) \times 10^{-11}$ A $\cdot$ m^3. Given a reported magnetic field gradient of 0.1 nT/cm, the RMQM interaction can introduce a non-negligible correction at the 10^{-14} precision level in the equivalence principle test. For the second sample, the RMQM components of interest are measured to be $43.8(4) \times 10^{-11}$ A $\cdot$ m^3. The uncertainty is mainly due to statistical error and the coil installation error. This RMQM measurement could be similarly applied to other gravitational experiments if needed.

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I. INTRODUCTION

In gravitational experiments, such as the measurement of the gravitational constant [1,2], the verification of the inverse square law [3,4], the test of the equivalence principle [5–7], and the detection of the gravitational waves [8,9], the signal of interest can easily be mimicked or contaminated by nongravitational couplings, particularly magnetic coupling. Despite efforts to minimize magnetic influences through attenuation or shielding, magnetic coupling requires careful evaluation due to its strong coupling nature and complex interaction mechanisms. Estimating the bias or disturbances introduced by magnetic coupling is thus a crucial issue.

The signals of interest in most gravitational experiments usually vary slowly over time, typically in the millihertz range. This allows magnetic coupling to be treated as a magnetostatic problem. To minimize magnetic coupling, test masses (TMs) are generally constructed from nonmagnetic materials, such as paramagnetic or diamagnetic substances. Nevertheless, ferromagnetic impurities are often introduced during mass fabrication. Therefore, the magnetic properties of the TMs can be characterized by their volume magnetic susceptibility, χ_{m}, for the nonmagnetic component, and by the remanent or residual magnetic magnetization, $\vec{M}_{\mathrm{re}}$, for the ferromagnetic fraction. These TMs interact with magnetic fields originating from the source masses or the environment. The interaction is characterized by the magnetic interaction energy [10,11]

$$U = -\int \left[\vec{M}_{\mathrm{re}}(\vec{r}) \cdot \vec{B}(\vec{r}) + \frac{1}{2}[\chi_{\mathrm{m}}\vec{H}(\vec{r})] \cdot \vec{B}(\vec{r})\right] \mathrm{d}V, \quad (1)$$

where the integral is over the entire TM volume, $\vec{B}(\vec{r})$ is the magnetic induction at position $\vec{r}$, and $\vec{H}(\vec{r})$ is the corresponding magnetic field strength. This interaction energy enables calculation of the mechanical force or torque, thus quantifying induced bias or noise.

Most gravitational experiments, however, focus only on the magnetic dipole interaction for the remanent term, often

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assuming higher-order contributions, such as the magnetic quadrupole interaction, to be negligible. For instance, in space-borne gravitational wave detection, the remanent magnetism of the TMs is typically characterized by the magnetic dipole moment $\vec{m} = \int \vec{M}_{\rm re} dV$ [12–15]. Conversely, in the torsion-balance test of the equivalence principle, the magnetic quadrupole moment is included to quantify the remanent magnetism of the TM [6,16]. Nevertheless, experimental measurement of the remanent magnetic quadrupole moment (RMQM) has rarely been reported. Approximating the remanent magnetic coupling without accounting for higher-order terms or experimental verification may be inadequate for modern gravitational experiments requiring stringent precision.

To address the magnetic quadrupole interaction in gravitational experiments, this paper focuses on measuring the RMQM of a macroscopic mass. We demonstrate this measurement using an artificial magnetic field in an equivalence principle test [6]. The proposed measurement techniques can also be adapted for other gravitational experiments.

II. MEASUREMENT PRINCIPLE

A. Remanent magnetic interaction

The magnetic interaction energy between the TM with remanent magnetization $\vec{M}_{\rm re}(\vec{r})$ and the magnetic field generated by the environment or nearby source masses can be expressed as

$$U_{\rm re} = -\int \vec{M}_{\rm re} \cdot \vec{B} dV, \tag{2}$$

where we neglect variations in the magnetic field $\vec{B}(\vec{r})$ within the TM volume due to the nonmagnetic material of the TM. For analytical simplicity, we express the magnetic field $\vec{B}(\vec{r})$ inside the TM volume using the magnetic scalar potential $\Psi(\vec{r})$, rather than the vector potential ([11], pp. 415),

$$\vec{B}(\vec{r}) = \begin{cases} -\mu_0 \nabla \Psi, & \vec{r} \notin V_s \\ -\mu_0 \nabla \Psi + \mu_0 \vec{M}_s(\vec{r}). & \vec{r} \in V_s \end{cases} \tag{3}$$

Here, μ_0 is the vacuum magnetic permeability. The gradient operator ∇ in a Cartesian coordinate system is given by $\nabla = \vec{e}_x \partial/\partial x + \vec{e}_y \partial/\partial y + \vec{e}_z \partial/\partial z$, with basis vectors $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$. $\vec{M}_s$ represents the magnetization distribution of the local magnetic source with volume V_s. The magnetic source is always located outside the TM region ($\vec{r} \notin V_s$).

Substituting Eq. (3) into the expression for $U_{\rm re}$ and applying the vector identity from Eq. (B3), we obtain

$$U_{\rm re} = \mu_0 \int \nabla \cdot (\Psi \vec{M}_{\rm re}) dV + \mu_0 \int \rho(\vec{r}) \Psi(\vec{r}) dV, \tag{4}$$

where $\rho(\vec{r}) = -\nabla \cdot \vec{M}_{\rm re}$. The first integral can be converted to a surface integral using Eq. (B6), and vanishes under the assumption of zero magnetization at the TM surface.

The scalar potential $\Psi(\vec{r})$ within the TM region, caused by the nearby source magnetization $\vec{M}_s(\vec{r}_s)$, is given by

$$\Psi(\vec{r}) = \frac{1}{4\pi} \int \frac{\rho_s(\vec{r}_s)}{|\vec{r} - \vec{r}_s|} dV_s + \frac{1}{4\pi} \oint \frac{\vec{M}_s(\vec{r}_{su}) \cdot d\vec{S}}{|\vec{r} - \vec{r}_{su}|}. \tag{5}$$

Here, $\rho_s(\vec{r}_s) = -\nabla_s \cdot \vec{M}_s(\vec{r}_s)$, where the gradient operator ∇_s applies to the magnetic source and is defined as $\nabla_s = \vec{e}_x \partial/\partial x_s + \vec{e}_y \partial/\partial y_s + \vec{e}_z \partial/\partial z_s$. $\vec{r}_{su}$ represents the position vector of a point on the external surface.

The form of the scalar potential Eq. (5) is analogous to the gravitational potential [17,18] and also includes the $1/|\vec{r} - \vec{r}_s|$ term, which can be expanded as a series of spherical harmonics [10,19]

$$\frac{1}{|\vec{r} - \vec{r}_s|} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} \mathcal{S}_{lm}(\vec{r}) \left[\frac{Y_{lm}(\vartheta_s, \varphi_s)}{r_s^{l+1}} \right], \tag{6}$$

for $|\vec{r}_s| > |\vec{r}|$. The function $\mathcal{S}_{lm}(\vec{r})$ is defined as

$$\mathcal{S}_{lm}(\vec{r}) = r^l Y^*_{lm}(\vartheta, \varphi), \tag{7}$$

where superscript "*" indicates the complex conjugate. The spherical coordinates of the vector $\vec{r}$ and $\vec{r}_s$ are (r, ϑ, φ) and $(r_s, \vartheta_s, \varphi_s)$, respectively. $Y_{lm}(\vartheta, \varphi)$ represents the spherical harmonics (see Refs. [10,19] or Appendix A).

Substituting Eq. (6) into Eq. (5), we obtain

$$\Psi(\vec{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} \mathcal{S}_{lm}(\vec{r}) Q_{lm}, \tag{8}$$

where the integrand in the surface integral of Eq. (5) has also been expanded using spherical harmonics. The term Q_{lm} corresponds to the integral over the magnetic source:

$$Q_{lm} = \int \rho_s(\vec{r}_s) \left[\frac{Y_{lm}(\vartheta_s, \varphi_s)}{r_s^{l+1}} \right] dV_s + \oint \left[\frac{Y_{lm}(\vartheta_{su}, \varphi_{su})}{r_{su}^{l+1}} \right] \vec{M}_s \cdot d\vec{S}. \tag{9}$$

This is referred to as the "magnetic multipole field," which includes uniform fields, field gradients and higher-order terms, as discussed in Appendix C. Substituting the expanded form of the scalar potential into Eq. (4) yields

$$U_{\mathrm{re}} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{\mu_0}{2l+1} q_{lm} Q_{lm}, \tag{10}$$

where q_{lm} is the volume integral over the TM

$$q_{lm} = \int \rho(\vec{r}) \mathcal{S}_{lm}(\vec{r}) \mathrm{d}V. \tag{11}$$

This is known as the magnetic multipole moment.

For $l = 0$ and $m = 0$, q_{00} can be expressed as a surface integral over the TM using Eq. (B6). Since the TM is assumed to have zero surface magnetization, $q_{00} = 0$.

B. Magnetic quadrupole moment

Magnetism is generally attributed to electric current. In this context, the remanent magnetism of the TM can be characterized by the current density $\vec{J}(\vec{r})$. In Cartesian coordinates, the magnetic multipole moment is defined in terms of the current density as [11,20]

$$\mu^{(l)}_{r_j r_k \cdots r_m} = \frac{l}{(l+1)!} \int (\vec{r} \times \vec{J})_j \underbrace{r_k \cdots r_m}_{l-1 \text{ terms}} \mathrm{d}V, \tag{12}$$

where l is a positive integer, and the indices $j, k, \ldots, m$ range over $\{1, 2, 3\}$. Here r_k represents the Cartesian components of the position vector $\vec{r} = (r_1, r_2, r_3) = (x, y, z)$.

For $l = 1$, the magnetic dipole moment is given by

$$\mu^{(1)}_{r_j} = \frac{1}{2} \int (\vec{r} \times \vec{J})_j \mathrm{d}V, \tag{13}$$

and for $l = 2$, the RMQM [21] is defined as

$$\mu^{(2)}_{r_j r_k} = \frac{1}{3} \int (\vec{r} \times \vec{j})_j r_k \mathrm{d}V. \tag{14}$$

As shown in Appendix D, the multipole moment defined by the current density is equivalent to the magnetization-based definition q_{lm}, owing to the fundamental relationship between current density and magnetization for the TM,

$$\vec{J}(\vec{r}) = \nabla \times \vec{M}_{\mathrm{re}}(\vec{r}). \tag{15}$$

Appendix D demonstrates that q_{10} and q_{11} can be expressed in terms of the dipole moment $\mu^{(1)}_{r_j}$ as

$$q_{10} = \frac{1}{2} \sqrt{\frac{3}{\pi}} \mu^{(1)}_z, \tag{16}$$

$$q_{11} = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \left[\mu^{(1)}_x - \mathrm{i} \mu^{(1)}_y \right], \tag{17}$$

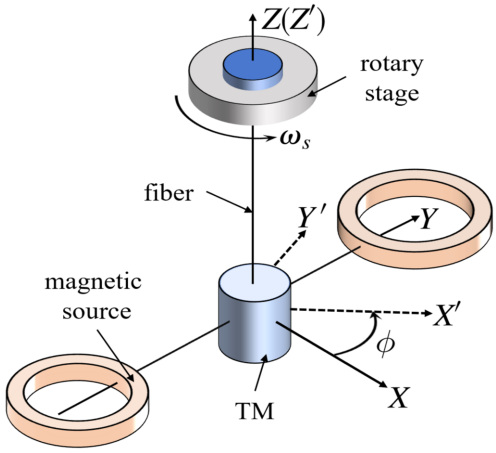


FIG. 1. The schematic diagram of the measurement instrument.

and q_{20}, q_{21} and q_{22} can be expressed in terms of the RMQM $\mu^{(2)}_{r_j r_k}$ as

$$q_{20} = \frac{3}{2} \sqrt{\frac{5}{\pi}} \mu^{(2)}_{zz}, \tag{18}$$

$$q_{21} = -\frac{1}{2} \sqrt{\frac{15}{2\pi}} \left[(\mu^{(2)}_{zx} + \mu^{(2)}_{xz}) - \mathrm{i}(\mu^{(2)}_{zy} + \mu^{(2)}_{yz}) \right], \tag{19}$$

$$q_{22} = \frac{1}{2} \sqrt{\frac{15}{2\pi}} \left[(\mu^{(2)}_{xx} - \mu^{(2)}_{yy}) - \mathrm{i}(\mu^{(2)}_{yx} + \mu^{(2)}_{xy}) \right]. \tag{20}$$

C. Torque due to remanent magnetic coupling

Figure 1 illustrates the schematic of our instrument. The TM is suspended by a fiber, allowing rotation around its axis. For clarity, we define two coordinate frames: the laboratory frame $O - XYZ$ and the TM-fixed frame $O' - X'Y'Z'$. Both origins coincide with the TM's center of mass, with the OZ and $O'Z'$ axes aligned along the fiber's direction. The OX, OY, $O'X'$ and $O'Y'$ axes lie in the horizontal plane. The azimuth angle ϕ of the TM is the angular deviation of $O'X'$ from OX.

To avoid confusion, the magnetic multipole moment in the $O' - X'Y'Z'$ frame is denoted as $\tilde{\mu}^{(l)}_{r_j r_k \cdots r_m}$, or $\tilde{q}_{lm}$. Considering the rotational relationship between the two frames, which transforms the spherical harmonics, we have

$$Y_{lm}(\vartheta, \varphi) = Y_{lm}(\vartheta', \varphi') e^{\mathrm{i}m\phi}. \tag{21}$$

The remanent magnetic interaction energy can then be written as

$$U_{\mathrm{re}} = \mu_0 \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} (\tilde{q}_{lm} e^{-\mathrm{i}m\phi}) Q_{lm}. \tag{22}$$

The minus sign in the exponent arises from the complex conjugate of the spherical harmonics. The mechanical torque along the OZ axis is given by [22,23]

$$\tau_z = -\frac{\partial U_{\mathrm{re}}}{\partial \phi}, \tag{23}$$

which can be expressed in matrix form as

$$\tau_z = \sum_{l=1}^{\infty}\sum_{m=1}^{l}\frac{-2m}{2l+1}\mu_0\begin{bmatrix}\mathrm{Re}[Q_{lm}] & \mathrm{Im}[Q_{lm}]\end{bmatrix} \cdot \begin{bmatrix}\mathrm{Re}[\tilde{q}_{lm}] & \mathrm{Im}[\tilde{q}_{lm}] \\ -\mathrm{Im}[\tilde{q}_{lm}] & \mathrm{Re}[\tilde{q}_{lm}]\end{bmatrix}\begin{bmatrix}\sin(-m\phi) \\ \cos(-m\phi)\end{bmatrix}. \tag{24}$$

Here, $\mathrm{Re}[Q]$, $\mathrm{Im}[Q]$ denote the real and imaginary parts of the complex quantity Q, respectively. The summation over m is transformed from the range $m=-l$ to l to $m=1$ to l. For $m=0$, the torque vanishes. This indicates that certain magnetic multipole moments, such as $\tilde{\mu}_z^{(1)}$ and $\tilde{\mu}_{zz}^{(2)}$, do not affect the rotational degree of freedom of the TM and thus cannot be measured with this method.

D. Measurement equation

In the torsion-balance test of the equivalence principle, the entire apparatus is continuously rotated by a turntable at an angular frequency ω_s. Assuming the two frames, $O-XYZ$ and $O'-X'Y'Z'$ coincide at the initial time, the azimuth angle ϕ is given by $\phi=\omega_s t$. When the magnetic source is stationary in the laboratory frame, the torque τ_z becomes sinusoidal, as described by Eq. (24). The torque components at frequencies $1\omega_s$ and $2\omega_s$ contain the information about the RMQM.

The sinusoidal torque causes the TM, suspended by a fiber, to rotate around the fiber at an oscillatory angle $\Theta(t)$. This angle can be obtained from the equation of motion, which accounts for both the magnetic torque τ_z and the torque due to the fiber twist

$$\tilde{\Theta}(\omega) = \frac{\tilde{\tau}_z(\omega)}{k}\cdot G(\omega), \tag{25}$$

where $\tilde{\Theta}(\omega)$ and $\tilde{\tau}_z(\omega)$ represent the Fourier transforms of the angle and torque, respectively. Here, k is the torsional constant of the suspending fiber, and $G(\omega)$ is the response function of the TM in the frequency domain,

$$G(\omega) = \frac{\omega_0^2}{(\omega_0^2-\omega^2)+\mathrm{i}(\omega_0^2/Q)} = |G(\omega)|e^{-\mathrm{i}\psi(\omega)}. \tag{26}$$

The angular frequency ω takes the values $m\omega_s (m=1,2,\ldots,l)$. ω_0 is the free torsional frequency of the TM in a high vacuum, and Q is the system quality factor, which is dominated by internal damping [24].

Once the angle $\tilde{\Theta}(\omega)(\omega=1\omega_s, 2\omega_s)$, induced only by the RMQM, is extracted from the data recorded by a corotating autocollimator, we can express it as

$$\tilde{\Theta}(\omega) = \tilde{\Theta}_\omega^{\sin}\sin(-\omega t) + \tilde{\Theta}_\omega^{\cos}\cos(-\omega t), \tag{27}$$

where $\tilde{\Theta}_\omega^{\sin}$ and $\tilde{\Theta}_\omega^{\cos}$ are the sine and cosine components of the Fourier transform of the angle. The torque, $\tau_\omega = \tau_\omega^{\sin}\sin(-\omega t) + \tau_\omega^{\cos}\cos(-\omega t)$, generated by the RMQM can then be derived as

$$\begin{bmatrix}\tau_\omega^{\sin} \\ \tau_\omega^{\cos}\end{bmatrix} = \frac{k}{|G(-\omega)|}\begin{bmatrix}\cos[\psi(-\omega)] & -\sin[\psi(-\omega)] \\ \sin[\psi(-\omega)] & \cos[\psi(-\omega)]\end{bmatrix}\begin{bmatrix}\tilde{\Theta}_\omega^{\sin} \\ \tilde{\Theta}_\omega^{\cos}\end{bmatrix}. \tag{28}$$

From this, the RMQM can be deduced as

$$\begin{bmatrix}\mathrm{Re}[\tilde{q}_{2m}] \\ \mathrm{Im}[\tilde{q}_{2m}]\end{bmatrix} = \frac{-5/(m\mu_0)}{2|Q_{2m}|^2}\begin{bmatrix}\mathrm{Re}[Q_{2m}] & \mathrm{Im}[Q_{2m}] \\ -\mathrm{Im}[Q_{2m}] & \mathrm{Re}[Q_{2m}]\end{bmatrix}\begin{bmatrix}\tau_\omega^{\sin} \\ \tau_\omega^{\cos}\end{bmatrix}, \tag{29}$$

where $m=1,2$. This equation represents the measurement equation for the RMQM. The magnetic torque generated by q_{21} coupling results in a sinusoidal signal with frequency ω_s, while the q_{22} coupling torque has a frequency of $2\omega_s$. It should be noted that the magnetic couplings with the same m-value will generate angle signals at the same frequency, as indicated by Eq. (24) with $\phi=\omega_s t$.

III. EXPERIMENTAL MEASUREMENT

A. Measurement apparatus

The apparatus employed here is a rotating torsion pendulum designed to test the equivalence principle [6]. Two samples shown in Fig. 2 are used. Sample I is a modified chiral pendulum composed of eight quartz crystals and nine glass rods, designed similar to the one used in

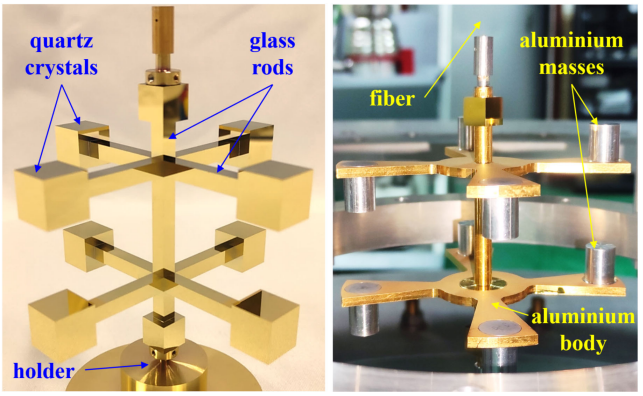


FIG. 2. Photos of sample I (left) and sample II (right). Sample I consisting of a pendulum with a single ferrule is mounted on a holder, while sample II is suspended by a fiber.

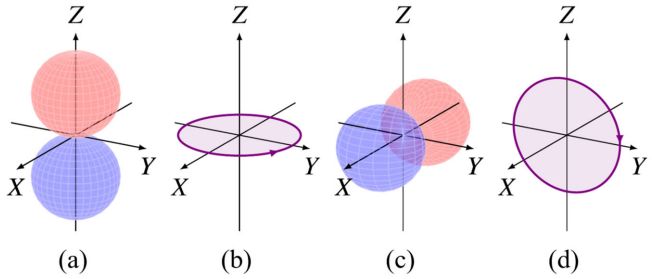


FIG. 3. The conceptual design of the current coils for Q_{10} and Q_{11} components. (a) and (c) show the functions of $|Y_{10}|\cos\varphi$ and $|Y_{11}|\cos\varphi$, respectively. (b) and (d) depict the coil designs for Q_{10} and Q_{11}, respectively.

earlier equivalence principle test. Sample II, primarily composed of aluminum alloy, is used to estimate the ambient gravitational gradient effects. To prevent electrostatic disturbances, each sample is coated with a thin metal layer after assembly. The masses of sample I and sample II are 72.0 g, and 67.9 g, respectively.

The samples are suspended sequentially using a tungsten fiber with a torsional constant of approximately 6.2×10^{-9} N m/rad. The free oscillation periods for the two samples are 700 s and 650 s, respectively. Each sample is placed in a chamber maintained at a pressure $\leq 5 \times 10^{-5}$ Pa. The vacuum is sustained by a corotating ion pump, ensuring a quality factor of $Q \approx 3 \times 10^3$. An autocollimator mounted outside the chamber measures the angular position of each sample. The chamber, including all attached components, is mounted below an air-bearing turntable that rotates it at a constant rate ω_s. During RMQM measurements, all magnetic shield layers [16] are removed. Environmental monitoring units record data simultaneously, including the temperature, and tilt of the rotary axis.

B. Design of current coils

Measuring the $\tilde{q}_{2m}$ component, as defined in Eq. (29), requires a magnetic field source with a known Q_{2m} component. To ensure accuracy, the magnetic source must minimize other components with the same m value as their coupling torques could mimic the desired signal.

The most widely used artificial magnetic field source is a current-carrying coil. Here, we utilize a planar round coil for simplicity. Designing coils to produce the specified field component requires identifying a configuration that generates only the desired component. Coil designs have been detailed in previous Refs. [25,26]. This paper focused on the conceptual design.

We first design configurations for the Q_{10} and Q_{11}, which serve as foundational models for Q_{21} and Q_{22}. The function $\mathrm{Re}[Y_{10}(\vartheta,\varphi)]$ is plotted to visualize the Q_{10} component, as shown in Fig. 3. The plot uses colors to indicate positive (red) and negative (blue) values. Assuming a magnet with locally uniform magnetization enclosed by a curved surface,

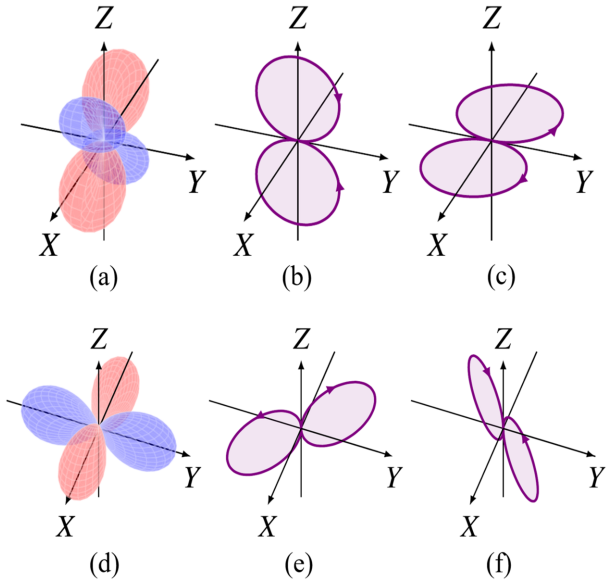


FIG. 4. The conceptual design of the current coils for Q_{21} and Q_{22} components. (a) and (d) show the functions of $|Y_{21}|\cos\varphi$ and $|Y_{22}|\cos\varphi$, respectively. (b) and (c) depict the coil design for Q_{21}, while (e) and (f) depict the design for Q_{22}.

we deduce that $\rho_s = -\nabla_s \cdot \vec{M}_s = 0$ within the magnet. This simplifies the analysis to the surface integral in Eq. (9). Regions with opposite integral signs correspond to distinct magnetic poles. The red region represents the magnetic north pole, while the blue region corresponds to the south pole. A symmetrical arrangement of magnetic poles generates a Q_{10} component equivalent to that of a single coil positioned in OXY plane, as illustrated in Fig. 3. This equivalence is not unique. This approach also applies to the Q_{11} coil configuration [Fig. 3(c)]. Similarly, coil configurations for Q_{21}, Q_{22} and other higher-order terms can be designed, as shown in Fig. 4.

C. Artificial magnetic field

The round coil has a diameter of $2r_0 \approx 1.0$ m and consists of approximately $N \approx 250$ turns of enameled wire with a diameter of about 2 mm. The coil's dimensions are $w = 6$ cm and $t = 2$ cm, as shown in Fig. 5. For the Q_{21} component, two coils are placed 2.0 m apart, forming the "Q_{21} configuration", as shown in Fig. 6(a). The coil configuration described in Sec. III B is not used directly for the Q_{22} component. As shown in Fig. 6(b), this modified arrangement is designed to produce a larger Q_{22} component. This arrangement is referred to as the Q_{22} configuration.

When both coils are powered by a current of I, we calculate the field components Q_{l1} and Q_{l2} for the two coil configurations using Eq. (9) with a specified magnetization. The coil magnetization is determined by the current density $j(\vec{r}_s)$, which is given by $j = NI/(wt)$. This determination is not unique, and we adopt the approach outlined in Ref. [27]:

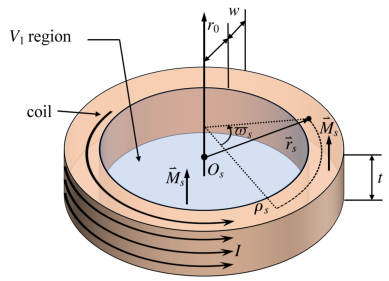


FIG. 5. A schematic drawing of a round coil with a constant current density. The only component of the current density is in the $\vec{\varpi}_s$ direction within the coil. The magnetization $\vec{M}_s$ is assumed to have only the axial component.

$$\vec{M}_s(\vec{r}_s) = \begin{cases} j \cdot w\vec{e}_{z_s}, & \vec{r}_s \in V_1 \\ j \cdot (r_0 + w - \rho_s)\vec{e}_{z_s}, & \text{inside coil} \\ \vec{0}. & \text{elsewhere} \end{cases} \tag{30}$$

Here, $\vec{r}_s = (\rho_s, \varpi_s, z_s)$ represents the position vector in cylindrical coordinates. This magnetization causes the integrand of the volume integral in Eq. (9) to be zero, leaving only the surface integral. Each Q_{lm} component can then be obtained. The Q_{lm} components produced by each configuration are listed in Tables I and II. The uncertainties primarily arise from an installation error of approximately 1 cm in the coil placement.

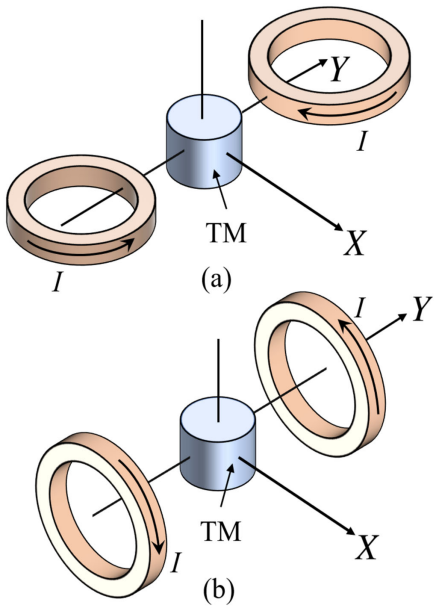


FIG. 6. The schematic diagram of coil configurations. (a) Q_{21} configuration: Coils are 2.0 m apart along the OY axis, with axes parallel to the OZ axis and currents flowing in opposite directions. (b) Q_{22} configuration: Coils are 1.0 m apart along the OY axis, with axes parallel to the OY axis and currents flowing in opposite directions.

TABLE I. Designed values of the magnetic field gradient Q_{lm} in Q_{21} coil configuration carrying current of 0.518 A.

	Q_{l1}	Q_{l2}	Unit
$l=1$	$0(0)+0(2)$i	$\cdots$	A/m^1
$l=2$	$0(1)+225(6)$i	$0(3)+0(0)$i	A/m^2
$l=3$	$0(0)+0(13)$i	$0(11)+0(4)$i	A/m^3
$l=4$	$0(0)-736(30)$i	$0(24)+0(0)$i	A/m^4

TABLE II. Designed values of the magnetic field gradient Q_{lm} in Q_{22} coil configuration carrying current of 0.518A.

	Q_{l1}	Q_{l2}	Unit
$l=1$	$0(4)+0(6)$i	$\cdots$	A/m^1
$l=2$	$0(0)+0(7)$i	$-686(10)+0(7)$i	A/m^2
$l=3$	$0(1)+0(4)$i	$0(4)+0(0)$i	A/m^3
$l=4$	$0(0)+0(10)$i	$221(12)+0(5)$i	A/m^4

To verify the designed field component, a three-axis flux gate magnetometer (Bartington Instruments, Mag-03 MSL100) was placed at the origin. To distinguish the coil field from the ambient field, the coils were driven by a square wave current. The magnetometer can measure both the field and its gradient, which are directly related to the Q_{lm} values, as described in Sec. C. The relationships are summarized as follows

$$Q_{11} = \frac{\sqrt{6\pi}}{\mu_0}[B_x|_{\vec{r}=\vec{0}} + \mathrm{i}B_y|_{\vec{r}=\vec{0}}], \tag{31}$$

$$Q_{21} = \frac{\sqrt{10\pi}}{\sqrt{3}\mu_0}[\partial_z B_x|_{\vec{r}=\vec{0}} + \mathrm{i}\partial_z B_y|_{\vec{r}=\vec{0}}], \tag{32}$$

$$Q_{22} = -\frac{\sqrt{10\pi}}{\sqrt{3}\mu_0}\left[\left.\frac{\partial_x B_x - \partial_y B_y}{2}\right|_{\vec{r}=\vec{0}} + \mathrm{i}\partial_x B_y|_{\vec{r}=\vec{0}}\right], \tag{33}$$

where $\partial_{r_i} B_j = \partial B_j / \partial r_i, (i,j = 1,2,3)$. The measured results are presented in Table III. The small value of Q_{11} is mainly due to the installation error in the coil positioning.

TABLE III. Designed and measured values of the magnetic field components for each coil configuration carrying current of 0.518A.

	Q_{21} configuration		Q_{22} configuration		
	Designed	Measured	Designed	Measured	Unit
B_x	0.0(0)	−0.01(1)	0(1)	0.2(1)	μT
B_y	0.0(6)	0.25(1)	0(2)	0.5(1)	μT
$\partial_z B_y$	0.87(2)	0.87(1)	0.00(0)	$\cdots$	μT/cm
B_Δ [a]	0.00(1)	$\cdots$	2.66(4)	2.7(1)	μT/cm

[a] $B_\Delta = (\partial_x B_x - \partial_y B_y)/2$.

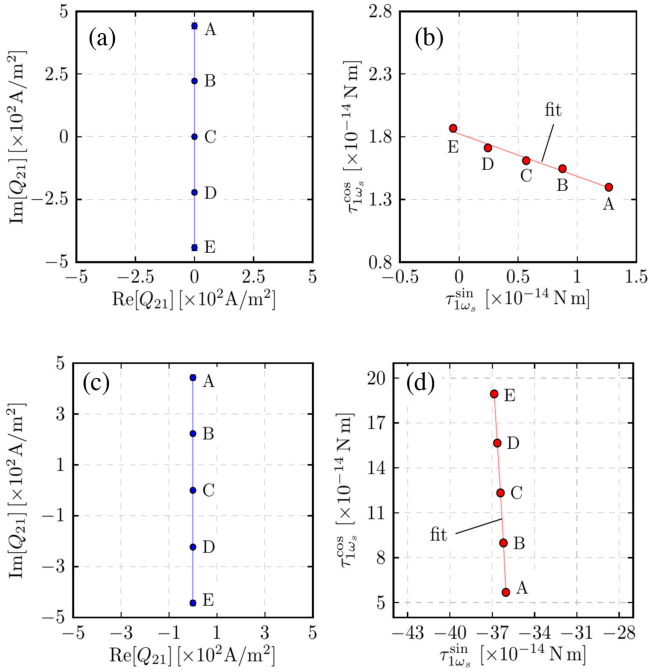


FIG. 7. Torque components at $1\omega_s$ varying with Q_{21}. (a) and (b) correspond to sample I, with currents A-E being 1.018, 0.512, 0.000, −0.512 and −1.018 A, respectively. (c) and (d) correspond to sample II, with currents A-E being 1.021, 0.514, 0.000, −0.514 and −1.021 A, respectively.

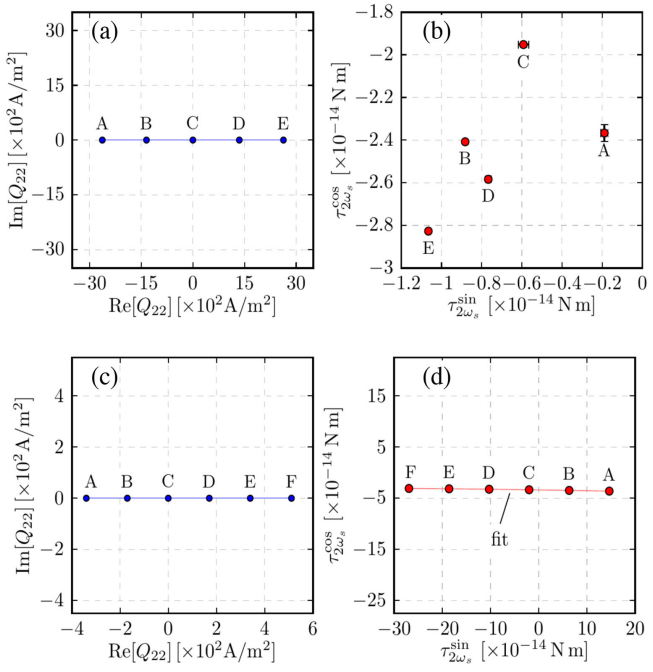


FIG. 8. Torque components at $2\omega_s$ varying with Q_{22}. (a) and (b) correspond to sample I, with currents A-E being 1.986, 1.018, 0.000, −1.018 and −1.986 A, respectively. (c) and (d) correspond to sample II, with currents A-F being 0.257, 0.128, 0.000, −0.128, −0.257 and −0.386 A, respectively.

D. Measurement of magnetic quadrupole moment

To measure RMQM component $\tilde{q}_{21}$ and $\tilde{q}_{22}$ for each sample, the Q_{21} and Q_{22} coil configurations were installed sequentially. For each coil configuration, the current is varied in defined steps. At each current step, data, including the TM angle and the tilt of the rotary axis, are collected.

The signal of interest is extracted using the correlation method. The data process procedure follows that described in Ref. [6]. First, a three-point filter is applied to eliminate the free torsion oscillations, followed by another three-point filter to remove the higher-order harmonics and polynomial drift. The filtered data are then divided into segments corresponding to integer multiples of rotation cycles. For each data segment, the correlation method is employed to extract the $\sin(-m\omega_s t)$ and $\cos(-m\omega_s t)$ components, where $m = 1,\ 2$. The tilt effect for $\sin(-\omega_s t)$ and $\cos(-\omega_s t)$ components, are corrected point-to-point [28]. The torque component can be obtained according to Eq. (28).

Figures 7 and 8 illustrate the corresponding torque components for each sample's response to the Q_{21} and Q_{22} coil configurations at each current step, respectively. The data points for the torque can be fitted using the least squares methods to obtain a straight line. The scale factor and angle between this line and another line corresponding to the multipole field can then be used to determine the RMQM components. The measured RMQM components, $\tilde{q}_{21}$ and $\tilde{q}_{22}$ for two samples are presented in the first row of Table IV.

TABLE IV. Error budgets for $\tilde{q}_{21}$ and $\tilde{q}_{22}$ measurement, unit: 10^{-11} A · m^3. Uncertainties are 1-σ.

	Sample I				Sample II			
	$\mathrm{Re}[\tilde{q}_{21}]$	$\mathrm{Im}[\tilde{q}_{21}]$	$\mathrm{Re}[\tilde{q}_{22}]$	$\mathrm{Im}[\tilde{q}_{22}]$	$\mathrm{Re}[\tilde{q}_{21}]$	$\mathrm{Im}[\tilde{q}_{21}]$	$\mathrm{Re}[\tilde{q}_{22}]$	$\mathrm{Im}[\tilde{q}_{22}]$
Measurement and statistical uncertainty	1.0(1)	2.9(1)	0.1(1)	0.1(1)	30.0(1)	1.9(1)	−48.48(3)	0.61(3)
Installation error of coils	0.0(0)	0.0(1)	0.0(0)	0.0(0)	0.0(8)	0.0(1)	0.0(7)	0.0(5)
Magnetic dipole	0.1(1)	0.2(0)	⋯	⋯	−3.8(0)	1.2(0)	⋯	⋯
Higher-order terms	∼ 0.01	∼ 0.01	< 0.01	< 0.01	∼ 0.08	∼ 0.08	∼ 0.01	∼ 0.01
Magnetic susceptibility	∼ 0.1	∼ 0.1	< 0.01	< 0.01	∼ 0.2	∼ 0.2	< 0.01	< 0.01
Total	0.9(2)	2.7(2)	0.1(1)	0.1(1)	33.8(8)	0.7(2)	−48.5(7)	0.6(5)

TABLE V. Magnetic dipole of two sample, unit: 10^{-9} A · m^2.

	Sample I	Sample II
$\mu_x^{(1)}$	−0.31(1)	16.6(1)
$\mu_y^{(1)}$	0.71(1)	5.7(1)

IV. SYSTEMATIC EFFECTS

A. Installation error of coils and magnetic dipole term

The installation errors, including those in coil position and attitude mounting, introduce uncertainties in the magnetic filed components of interest, and contribute to a small Q_{11}, as discussed in Sec. III C. For the measurement of $\tilde{q}_{21}$ or $\tilde{q}_{22}$ for each sample, the uncertainties in Q_{21} or Q_{22} are taken as the values with the maximum current. The induced uncertainties in $\tilde{q}_{21}$ or $\tilde{q}_{22}$ are presented in the second row of Table IV.

Since the torque due to the dipole $\tilde{q}_{11}$ of each sample, coupled by the small Q_{11}, can also introduce the same frequency as the $\tilde{q}_{21}$ interaction term, a correction must be applied. The magnetic dipole components of each sample were measured using the method described in Ref. [16], and the results are listed in Table V. The corrections due to the dipole are presented in the third row of Table IV. The small values of Q_{11}, coupled with $\tilde{q}_{11}$, do not affect the measurement of $\tilde{q}_{22}$, as the induced frequencies are different.

B. Magnetic higher-order terms

According to Eq. (24), higher-order terms with $l \geq 3$ may be considered. In fact, the series in Eq. (24) converges with a factor of r_t/R_t, where r_t and R_t are typical values for the sample dimension and the distance between the magnetic source and the sample. Here, r_t for both samples is approximately 0.05 m, and $R_t \sim 0.5$ m for both coil configurations, yielding $r_t/R_t \leq 0.1$. Therefore, the items to consider are the several leading ones, $\tilde{q}_{3m}Q_{3m}$ and $\tilde{q}_{4m}Q_{4m}$ couplings with $m = 1, 2$. To provide an upper bound on these terms, we estimate the magnitudes of $\tilde{q}_{3m}$ and $\tilde{q}_{4m}$ for each samples. Considering the definition of $\tilde{q}_{lm}$ in Eq. (11) or Eq. (E1), we can infer that

$$|\tilde{q}_{l1}| = \left| \int \vec{M}_{\mathrm{re}} \cdot \nabla[r^l Y_{l1}^*] \mathrm{d}V \right| \sim r_t^{l-1} |\tilde{q}_{11}|, \tag{34}$$

for the $m = 1$ terms, and

$$|\tilde{q}_{l2}| = \left| \int \vec{M}_{\mathrm{re}} \cdot \nabla[r^l Y_{l2}^*] \mathrm{d}V \right| \sim r_t^{l-2} |\tilde{q}_{22}|, \tag{35}$$

for the $m = 2$ terms. Thus, the $\tilde{q}_{31}$ for the two samples can be estimated to have magnitudes of 2×10^{-12} A · m^4 and 4×10^{-11} A · m^4, respectively, and $|\tilde{q}_{41}| \sim 1 \times 10^{-13}$ A m^5, and 2×10^{-13} A m^5 for two samples, respectively. By substituting the magnetic multipole field for the Q_{21} coil configuration, we can deduce the magnitude of torque of the two samples. The $\tilde{q}_{32}Q_{32}$ and $\tilde{q}_{42}Q_{42}$ terms can be estimated in a similar manner, with estimated values of $|\tilde{q}_{22}| \sim 10^{-12}$ and $\sim 5 \times 10^{-10}$ A m^3 for two samples respectively. The maximum magnitudes of the torque are taken as the maximum estimated uncertainties for the $\tilde{q}_{21}$ and $\tilde{q}_{22}$ measurement of each samples, respectively. The corresponding corrections are presented in the fourth row in Table IV.

C. Magnetic susceptibility

The magnetization susceptibility χ_{m} of the sample material is assumed to be both linear and isotropic. The magnetic field strength inside the TM volume is given by

$$\vec{H}(\vec{r}) = \frac{1}{\mu} \vec{B}(\vec{r}), \tag{36}$$

where $\mu = \mu_0(1 + \chi_{\mathrm{m}})$ is the permeability of the medium, assuming no changes in $\vec{B}(\vec{r})$ due to remanent magnetism. The magnetic interaction energy due to susceptibility can be expressed as

$$U_{\mathrm{sus}} = -\frac{1}{2} \int \frac{\chi_{\mathrm{m}}}{\mu_0(1 + \chi_{\mathrm{m}})} \vec{B}(\vec{r}) \cdot \vec{B}(\vec{r}) \mathrm{d}V. \tag{37}$$

Substituting Eq. (3) into the above expression yields

$$U_{\mathrm{sus}} = -\frac{1}{2} \frac{\mu_0 \chi_{\mathrm{m}}}{1 + \chi_{\mathrm{m}}} \int (\nabla \Psi) \cdot (\nabla \Psi) \mathrm{d}V. \tag{38}$$

The magnetic scalar potential can be expressed using the magnetic multipole field Q_{lm}, as shown in Eq. (C1).

Upon continuous rotation of the sample, the magnetic field, $\vec{B}(\vec{r}) = -\mu_0 \nabla \Psi$, can be decomposed into two components. The first component is the invariant part $\vec{B}^{\mathrm{co}}(\vec{r})$ in the corotating frame $O' - X'Y'Z'$, given by

$$\vec{B}^{\mathrm{co}}(\vec{r}) = -\mu_0 \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} Q_{lm}^{\mathrm{co}} \nabla[\mathcal{S}_{lm}(\vec{r})], \tag{39}$$

where Q_{lm}^{co} primarily originates from the ion pump corotating with the turntable. The second component is the invariant part, $\vec{B}^{\mathrm{lab}}(\vec{r})$, in the lab frame $O - XYZ$, rotated about the axis OZ in the $O' - X'Y'Z'$ frame, described by

$$\vec{B}^{\mathrm{lab}}(\vec{r}) = -\mu_0 \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{1}{2l+1} Q_{lm}^{\mathrm{lab}} \nabla[\mathcal{S}_{lm}(\vec{r})]. \tag{40}$$

The relationship between the basis vectors of the two frames is expressed as

$$\begin{bmatrix} \vec{e}_x \\ \vec{e}_y \\ \vec{e}_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{e}_{x'} \\ \vec{e}_{y'} \\ \vec{e}_{z'} \end{bmatrix}. \quad (41)$$

This leads to the transformation of the spherical harmonics between the two frames,

$$\mathcal{S}_{lm} = \tilde{\mathcal{S}}_{lm} e^{-\mathrm{i}m\phi}, \qquad Q^{\mathrm{co}}_{lm} = \tilde{Q}^{\mathrm{co}}_{lm} e^{\mathrm{i}m\phi}, \quad (42)$$

and the gradient operator in the two frames

$$\nabla = \sum_{i=1}^{3} \vec{e}_{r_i} \frac{\partial}{\partial r_i} = \sum_{i=1}^{3} \vec{e}_{r'_i} \frac{\partial}{\partial r'_i} = \nabla'. \quad (43)$$

By substituting Eqs. (39)–(43), the susceptibility-induced energy is derived as

$$U_{\mathrm{sus}} = -\frac{1}{2}\frac{\mu_0\chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} \int \left[\sum_{l,m} \frac{\nabla'[\tilde{\mathcal{S}}_{lm}]}{2l+1} (\tilde{Q}^{\mathrm{co}}_{lm} + Q^{\mathrm{lab}}_{lm} e^{-\mathrm{i}m\phi})\right] \cdot \left[\sum_{l_2,m_2} \frac{\nabla'[\tilde{\mathcal{S}}_{l_2m_2}]}{2l_2+1} (\tilde{Q}^{\mathrm{co}}_{l_2m_2} + Q^{\mathrm{lab}}_{l_2m_2} e^{-\mathrm{i}m_2\phi})\right] \mathrm{d}V. \quad (44)$$

The torque due to susceptibility can be calculated as

$$\tau_{\mathrm{sus},z} = -\frac{\partial U_{\mathrm{sus}}}{\partial\phi}, \quad (45)$$

yielding

$$\tau_{\mathrm{sus},z} = \frac{\mu_0\chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} \int \left[\sum_{l,m} \frac{\nabla'[\tilde{\mathcal{S}}_{lm}]}{2l+1} (-\mathrm{i}m Q^{\mathrm{lab}}_{lm} e^{-\mathrm{i}m\phi})\right] \cdot \left[\sum_{l_2,m_2} \frac{\nabla'[\tilde{\mathcal{S}}_{l_2m_2}]}{2l_2+1} (\tilde{Q}^{\mathrm{co}}_{l_2m_2} + Q^{\mathrm{lab}}_{l_2m_2} e^{-\mathrm{i}m_2\phi})\right] \mathrm{d}V. \quad (46)$$

Appendix F presents the leading terms for $1\omega_s$ and $2\omega_s$ torques, given by

$$\tau_{1\omega_s} = \frac{\mu_0\chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} \frac{\pi}{12} V(\mathrm{i}Q^{\mathrm{lab}}_{11}\tilde{Q}^{\mathrm{co}}_{1,-1} e^{-\mathrm{i}\omega_s t} - \mathrm{i}Q^{\mathrm{lab}}_{1,-1}\tilde{Q}^{\mathrm{co}}_{11} e^{\mathrm{i}\omega_s t}), \quad (47)$$

and

$$\tau_{2\omega_s} = -\frac{2\mu_0\chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} (\mathrm{i}\tilde{\mathcal{Q}} Q^{\mathrm{lab}}_{22}\tilde{Q}^{\mathrm{co}}_{1,-1} e^{-\mathrm{i}2\omega_s t} + \mathrm{i}\tilde{\mathcal{Q}}^{\dagger} Q^{\mathrm{lab}}_{2,-2}\tilde{Q}^{\mathrm{co}}_{11} e^{\mathrm{i}2\omega_s t}). \quad (48)$$

respectively, where V is the sample volume. $\tilde{\mathcal{Q}}$ is defined as

$$\tilde{\mathcal{Q}} = \frac{\sqrt{5}}{20\pi} \int (x' - \mathrm{i}y')\mathrm{d}V, \quad (49)$$

which characterizes the geometric symmetry of the sample.

To evaluate the impact of each sample on the $\tilde{q}_{21}$ measurement, we substitute the parameters in Eq. (46). The volumes of the two samples are approximately 2.9×10^{-5} m^3, and 2.5×10^{-5} m^3, respectively. The magnetic susceptibilities χ_{m} for the two samples are estimated to be 1.4×10^{-5} [29] and 2.2×10^{-5} [30], respectively. The magnitudes of $\tilde{Q}^{\mathrm{co}}_{1,-1}$ and $\tilde{Q}^{\mathrm{co}}_{11}$ are measured to be approximately 8.6 A/m [16], while the maximum magnitudes of Q^{lab}_{11} and $Q^{\mathrm{lab}}_{1,-1}$ are approximately 1.7 A. The maximum magnitude of $1\omega_s$ torque is estimated to be approximately 3×10^{-16} N m. The uncertainty in the $\tilde{q}_{21}$ measurement is taken as the maximum uncertainty for each $\tilde{q}_{21}$ component, as shown in the fifth row in Table IV. Similar estimations are made for the $\tilde{q}_{22}$ measurement, with $\mathcal{Q} \sim 3.7 \times 10^{-11}$ m^4 both for two samples.

V. CONCLUSION AND DISCUSSION

To investigate the RMQM interaction in gravitational experiments, we measured the RMQM components of the TM. We used two macroscopic masses as samples: one optimized quartz crystal pendulum and one aluminum alloy pendulum for the equivalence principle test. The measurement instrument consists of a rotating torsion pendulum with custom-designed magnetic coils. Different coil configurations were designed to measure each RMQM components. The measured results of the RMQM components for two samples are $\tilde{q}_{21} = [0.9(2) + \mathrm{i}2.7(2)] \times 10^{-11}$ A m^3,$[33.8(8) + \mathrm{i}0.7(2)] \times 10^{-11}$ A m^3, and $\tilde{q}_{22} = [0.1(1) + \mathrm{i}0.1(1)] \times 10^{-11}$ A m^3, $[-48.5(7) + \mathrm{i}0.6(5)] \times 10^{-11}$ A m^3, respectively. These results correspond to the components in the Cartesian coordinate system: $(\tilde{\mu}^{(2)}_{zx} + \tilde{\mu}^{(2)}_{xz}) = -1.2(3) \times 10^{-11}$ A · m^3, $-43.8(4) \times 10^{-11}$ A · m^3, $(\tilde{\mu}^{(2)}_{zy} + \tilde{\mu}^{(2)}_{yz}) = 3.5(3) \times 10^{-11}$ A · m^3, $0.9(3) \times 10^{-11}$ A · m^3, $(\tilde{\mu}^{(2)}_{xx} - \tilde{\mu}^{(2)}_{yy}) = 0.1(1) \times 10^{-11}$ A · m^3,$-62.8(9) \times 10^{-11}$ A · m^3, and $(\tilde{\mu}^{(2)}_{yx} + \tilde{\mu}^{(2)}_{xy}) = 0.1(1) \times 10^{-11}$ A · m^3,$0.8(6) \times 10^{-11}$ A · m^3. The uncertainties are mainly due to the statistical errors and the coil installation errors.

For the equivalence principle test, aiming for a precision level of 10^{-14}, if the TMs have a RMQM with a magnitude of $[(\tilde{\mu}^{(2)}_{zx} + \tilde{\mu}^{(2)}_{xz})^2 + (\tilde{\mu}^{(2)}_{zy} + \tilde{\mu}^{(2)}_{yz})^2]^{1/2} \approx 3.7 \times 10^{-11}$ A · m^3, the RMQM interaction could introduce a torque of 3.6×10^{-19} N m, given a reported magnetic field gradient of 0.1 nT/cm [16]. This torque can mimic an equivalence-principle violation signal of approximately 4×10^{-14}, which is non-negligible. It is not advisable to reduce the pendulum RMQM, as it is primarily determined by the magnetization distribution. Unlike mass distribution in gravitation [18],

which is easier to control, the magnetization distribution presents greater challenges. Therefore, to minimize the RMQM coupling, the solution is to reduce the magnetic field gradient. Since reducing the magnetic field gradient by adding coils with an appropriate configuration may introduce additional temperature disturbances, improving magnetic shielding is a preferable approach. Efforts are underway to optimize the magnetic shielding by at least a factor of 4 for the equivalence principle test.

For other gravitational experiments, such as the measurements of the gravitational constant or space-borne detection of gravitational wave, the RMQM of the TM could be roughly evaluated using Eq. (34). Once the magnetic field gradient is determined, the RMQM interaction can be estimated. If the correction due to the RMQM interaction is required, the RMQM of the TMs can be measured with the proposed coil configuration.

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DATA AVAILABILITY

The data supporting this study's findings are available within the article.

APPENDIX A: SPHERICAL HARMONICS

The spherical harmonics are defined as [10,19]

$$Y_{lm}(\vartheta,\varphi)=\left[\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}\right]^{1/2}P_l^m(\cos\vartheta)e^{im\varphi},$$

where $P_l^m(\cos\vartheta)$ are the associated Legendre polynomials, defined as

$$P_l^m(\cos\vartheta)=(-1)^m\frac{(1-\cos^2\theta)^{m/2}}{2^l l!}\frac{\mathrm{d}^{l+m}(\cos^2\vartheta-1)^l}{\mathrm{d}(\cos\vartheta)^{l+m}}.$$

The integer l takes the values 0,1,2,$\cdots$ and for each l, integer m ranges from $-l$ to l.

APPENDIX B: VECTOR IDENTITY AND INTEGRAL IDENTITY

The subsequent derivation requires several vector identities, including the gradient of the product of two scalars

$$\nabla(uv)=v\nabla u+u\nabla v, \tag{B1}$$

the curl and divergence of a scalar-vector product

$$\nabla\times(u\vec{A})=u\nabla\times\vec{A}+\nabla u\times\vec{A}, \tag{B2}$$

$$\nabla\cdot(u\vec{A})=u\nabla\cdot\vec{A}+\nabla u\cdot\vec{A}, \tag{B3}$$

and the gradient of the dot product of two vectors

$$\nabla(\vec{A}\cdot\vec{B})=\vec{A}\times(\nabla\times\vec{B})+(\vec{A}\cdot\nabla)\vec{B}+\vec{B}\times(\nabla\times\vec{A})+(\vec{B}\cdot\nabla)\vec{A}. \tag{B4}$$

Additionally, the curl of the cross product of two vectors is required

$$\nabla\times(\vec{A}\times\vec{B})=(\vec{B}\cdot\nabla)\vec{A}-(\vec{A}\cdot\nabla)\vec{B}-\vec{B}(\nabla\cdot\vec{A})+\vec{A}(\nabla\cdot\vec{B}). \tag{B5}$$

Several integral identities can also be applied, including the divergence theorem

$$\int_V\nabla\cdot\vec{A}\mathrm{d}V=\int_S\vec{A}\cdot\mathrm{d}\vec{S}, \tag{B6}$$

the integral of the gradient

$$\int_V(\nabla u)\mathrm{d}V=\int_S u\mathrm{d}\vec{S}, \tag{B7}$$

and the integral of the curl

$$\int_V\nabla\times\vec{A}\mathrm{d}V=\int_S\mathrm{d}\vec{S}\times\vec{A}. \tag{B8}$$

APPENDIX C: MAGNETIC MULTIPOLE FIELD

As previously discussed in Sec. II A, the magnetic scalar potential $\Psi(\vec{r})$ within the TM region can be represented in spherical coordinates as

$$\Psi(\vec{r})=\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\frac{1}{2l+1}\mathcal{S}_{lm}(\vec{r})Q_{lm}. \tag{C1}$$

Alternatively, $\Psi(\vec{r})$ can be approximated using a Taylor expansion in Cartesian coordinates

$$\Psi(\vec{r})=\Psi(\vec{0})+\vec{r}\cdot(\nabla\Psi)|_{\vec{r}=\vec{0}}+\left[\frac{1}{2!}(\vec{r}\cdot\nabla)^2\Psi|_{\vec{r}=\vec{0}}\right]+\text{higher-order terms}. \tag{C2}$$

By comparing Eq. (C1) with Eq. (C2), we find that the multipole field component Q_{1m} ($m=0,1$) corresponds to the term

$$\Psi_{r_i}=\left.\frac{\partial\Psi}{\partial r_i}\right|_{\vec{r}=0},\qquad(i=1,2,3) \tag{C3}$$

and Q_{2m} $(m = 0, 1, 2)$ corresponds to the term

$$\Psi_{r_i r_j} = \left.\frac{\partial^2 \Psi}{\partial r_i \partial r_j}\right|_{\vec{r}=\vec{0}}. \quad (i, j = 1, 2, 3) \tag{C4}$$

This correspondence is based on the powers of r in Eq. (C1) and Eq. (C2).

From Eq. (3), the magnetic field $\vec{B}(\vec{r})$ outside the magnetic source is given by

$$B_i(\vec{r}) = -\mu_0 \Psi_{r_i} - \mu_0 \sum_{j=1}^{3} r_j \Psi_{r_j r_i} + \cdots. \tag{C5}$$

From this equation, we identify Ψ_{r_i} as the uniform component, and $\Psi_{r_i r_j}$ as the field gradient component.

By substituting Eq. (5) and inserting the spherical harmonics in the Cartesian system into the definition Eq. (9), we can establish the relationships between the multipole field Q_{lm} and the uniform field, field gradient or high order field components [22]. For example, for $l = 1$, we have

$$Q_{10} = 2\sqrt{3\pi}\Psi_z, \tag{C6}$$

$$Q_{11} = -\sqrt{6\pi}(\Psi_x + \mathrm{i}\Psi_y). \tag{C7}$$

For $l = 2$, we have

$$Q_{20} = \sqrt{5\pi}\Psi_{zz}, \tag{C8}$$

$$Q_{21} = -\sqrt{\frac{10\pi}{3}}(\Psi_{zx} + \mathrm{i}\Psi_{zy}), \tag{C9}$$

$$Q_{22} = \sqrt{\frac{10\pi}{3}}\left(\frac{\Psi_{xx} - \Psi_{yy}}{2} + \mathrm{i}\Psi_{xy}\right). \tag{C10}$$

APPENDIX D: MAGNETIC MULTIPOLE MOMENT IN CARTESIAN SYSTEM

The magnetic multipole moment in a Cartesian coordinate system is defined by Eq. (12). The integrand involves two functions: a vector function and a scalar function. The vector function is defined as

$$\vec{F} = \vec{r} \times (\nabla \times \vec{M}_{\mathrm{re}}), \tag{D1}$$

and the scalar function is given by

$$g(r_k, \ldots, r_m) = \underbrace{r_k \cdots r_m}_{l\text{-1 terms}}. \tag{D2}$$

The magnetic multipole moment, as defined in Eq. (12), can now be expressed as

$$\mu^{(l)}_{r_j r_k \cdots r_m} = \frac{l}{(l+1)!}\int \vec{F}_j \cdot g \mathrm{d}V. \tag{D3}$$

For $l = 1$, the function g simplifies to $g = 1$.

Applying the vector identity from Eq. (B4) (with $\vec{A} = \vec{r}$ and $\vec{B} = \vec{M}_{\mathrm{re}}$), we can rewrite $\vec{F}$ as

$$\vec{F} = \nabla(\vec{r} \cdot \vec{M}_{\mathrm{re}}) - (\vec{r} \cdot \nabla)\vec{M}_{\mathrm{re}} - (\vec{M}_{\mathrm{re}} \cdot \nabla)\vec{r}. \tag{D4}$$

We used the identity $\nabla \times \vec{r} = \vec{0}$. The last term on the right-hand side simplifies as follows:

$$(\vec{M}_{\mathrm{re}} \cdot \nabla)\vec{r} = \sum_{k=1}^{3} M_{\mathrm{re},r_k} \partial\vec{r}/\partial r_k = \vec{M}_{\mathrm{re}}. \tag{D5}$$

Next, applying the vector identity from Eq. (B5) (with $\vec{A} = \vec{r}$ and $\vec{B} = \vec{M}_{\mathrm{re}}$), we transform the second term

$$(\vec{r} \cdot \nabla)\vec{M}_{\mathrm{re}} = -\nabla \times (\vec{r} \times \vec{M}_{\mathrm{re}}) - 2\vec{M}_{\mathrm{re}} + \vec{r}(\nabla \cdot \vec{M}_{\mathrm{re}}). \tag{D6}$$

Here, we used Eq. (D5) and the identify $\nabla \cdot \vec{r} = 3$.

Substituting Eqs. (D6) and (D5) gives

$$\vec{F} = \nabla(\vec{r} \cdot \vec{M}_{\mathrm{re}}) + \nabla \times (\vec{r} \times \vec{M}_{\mathrm{re}}) + \vec{M}_{\mathrm{re}} - \vec{r}(\nabla \cdot \vec{M}_{\mathrm{re}}). \tag{D7}$$

Next, we consider the function $\vec{F} \cdot g$ in the integrand of Eq. (D3) and proceed with further transformations. The product of g and the first term on the right-hand side of Eq. (D7) can be rewritten as

$$g \cdot \nabla(\vec{r} \cdot \vec{M}_{\mathrm{re}}) = \nabla(g\vec{r} \cdot \vec{M}_{\mathrm{re}}) - \vec{r} \cdot \vec{M}_{\mathrm{re}} \nabla g,$$

using the vector identity from Eq. (B1) (with $u = g$ and $v = \vec{r} \cdot \vec{M}_{\mathrm{re}}$). Consequently, the product of g and the second term in Eq. (D7) becomes

$$g\nabla \times (\vec{r} \times \vec{M}_{\mathrm{re}}) = \nabla \times (g\vec{r} \times \vec{M}_{\mathrm{re}}) - \nabla g \times (\vec{r} \times \vec{M}_{\mathrm{re}}), \tag{D8}$$

using the vector identity from Eq. (B2) (with $u = g$ and $\vec{A} = \vec{r} \times \vec{M}_{\mathrm{re}}$).

Therefore, the integrand in Eq. (D3) can be written as

$$\begin{aligned}\vec{F}_j \cdot g = {} & [\nabla(g\vec{r} \cdot \vec{M}_{\mathrm{re}})]_j - \vec{r} \cdot \vec{M}_{\mathrm{re}}[\nabla g]_j + [\nabla \times (g\vec{r} \times \vec{M}_{\mathrm{re}})]_j \\ & - [\nabla g \times (\vec{r} \times \vec{M}_{\mathrm{re}})]_j + g[\vec{M}_{\mathrm{re}}]_j - (\nabla \cdot \vec{M}_{\mathrm{re}}) r_j. \end{aligned} \tag{D9}$$

The integral of the first term on the right-hand side of Eq. (D9) can be converted to a surface integral,

$$\int [\nabla(g\vec{r} \cdot \vec{M}_{\mathrm{re}})]_j \mathrm{d}V = \left[\int g_S \vec{r}_S \cdot \vec{M}_{\mathrm{re},S} \mathrm{d}\vec{S}\right]_j = 0,$$

where the subscript S denotes the function at the outside surface of the TM, and the surface magnetization $\vec{M}_{\mathrm{re,S}}$ of the TM is assumed to be zero. Similarly, the integral of the third term can also be converted to a surface integral,

$$\int \nabla \times (g\vec{r} \times \vec{M}_{\mathrm{re}})_j \mathrm{d}V = \left[\int \mathrm{d}\vec{S} \times (g_S \vec{r}_S \times \vec{M}_{\mathrm{re},S})\right]_j = 0.$$

By substituting the integral identity from Eq. (B3) into the integral for the last term, we obtain

$$\begin{aligned}\int \nabla \cdot \vec{M}_{\mathrm{re}}[\vec{r}]_j \mathrm{d}V &= \int [\nabla \cdot (r_j \vec{M}_{\mathrm{re}}) - \nabla r_j \cdot \vec{M}_{\mathrm{re}}]\mathrm{d}V \\ &= \int \mathrm{d}\vec{S} \cdot (r_{j,S}\vec{M}_{\mathrm{re},S}) - \int \nabla r_j \cdot \vec{M}_{\mathrm{re}} \mathrm{d}V,\end{aligned} \tag{D10}$$

where the integral identity Eq. (B6) is used, and the surface integral vanishes for $\vec{M}_{\mathrm{re},S} = \vec{0}$.

Consequently, the integrals of the four remaining terms in Eq. (D9) yield the multipole moment

$$\begin{aligned}\mu^{(l)}_{r_j r_k \cdots r_m} = \frac{l}{(l+1)!}\int [-\vec{r} \cdot \vec{M}_{\mathrm{re}}[\nabla g]_j - [\nabla g \times (\vec{r} \times \vec{M}_{\mathrm{re}})]_j \\ + g[\vec{M}_{\mathrm{re}}]_j + (\nabla r_j \cdot \vec{M}_{\mathrm{re}})]\mathrm{d}V.\end{aligned} \tag{D11}$$

Finally, for $l = 1$, the magnetic dipole moment is derived from Eq. (D11) as

$$\mu^{(1)}_{r_i} = \int M_{\mathrm{re},r_i}\mathrm{d}V \quad (i = 1, 2, 3). \tag{D12}$$

Similarly, for $l = 2$, the reduced RMQM $(i, j = 1, 2, 3)$ is derived as

$$\mu^{(2)}_{r_i r_j} = \begin{cases} -\frac{1}{3}\int \left(\sum_{k=1}^{3} r_k M_{\mathrm{re},r_k} - 3 r_i M_{\mathrm{re},r_i}\right)\mathrm{d}V, & i = j \\ \int r_j M_{\mathrm{re},r_i}\mathrm{d}V, & i \neq j. \end{cases} \tag{D13}$$

Higher-order multipole moments can similarly be derived from Eq. (D11).

APPENDIX E: MAGNETIC MULTIPOLE MOMENT IN SPHERICAL COORDINATE SYSTEM

Equation (11) defines the magnetic multipole moment in spherical coordinates. By substituting the identity Eq. (B3), assuming zero surface magnetization of the sample material, Eq. (11) can be rewritten as

$$q_{lm} = \int \vec{M}_{\mathrm{re}} \cdot \nabla[r^l Y^{\dagger}_{lm}(\vartheta, \varphi)]\mathrm{d}V. \tag{E1}$$

The spherical harmonic functions in Cartesian coordinates allow the derivation of specific terms as follows. For the dipole terms ($l = 1$), the following expressions hold

$$q_{10} = \frac{1}{2}\sqrt{\frac{3}{\pi}}\left[\int M_{\mathrm{re},z}\mathrm{d}V\right] \tag{E2}$$

$$q_{11} = -\frac{1}{2}\sqrt{\frac{3}{2\pi}}\left[\int M_{\mathrm{re},x}\mathrm{d}V - \mathrm{i}\int M_{\mathrm{re},y}\mathrm{d}V\right]. \tag{E3}$$

For the quadrupole terms ($l = 2$), the following expressions hold

$$q_{20} = -\frac{1}{2}\sqrt{\frac{5}{\pi}}\int (xM_x + yM_y - 2zM_z)\mathrm{d}V, \tag{E4}$$

$$q_{21} = -\frac{1}{2}\sqrt{\frac{15}{2\pi}}\int [(zM_x + xM_z) - \mathrm{i}(zM_y + yM_z)]\mathrm{d}V, \tag{E5}$$

$$q_{22} = \frac{1}{2}\sqrt{\frac{15}{2\pi}}\int [(xM_x - yM_y) - \mathrm{i}(yM_x + xM_y)]\mathrm{d}V. \tag{E6}$$

By comparing the dipole expressions in the Cartesian system Eq. (D12), we can establish their relationship, as shown in Eqs. (16) and (17). Similarly, the relationship between the magnetic quadrupole terms in two coordinate systems can be established, as shown in Eqs. (18)–(20).

APPENDIX F: TORQUE DUE TO MAGNETIC SUSCEPTIBILITY

The torque given by Eq. (46) can be further expanded as

$$\begin{aligned}\tau_{\mathrm{sus},z} = \frac{\mu_0 \chi_{\mathrm{m}}}{1 + \chi_{\mathrm{m}}}\Bigg[\sum_{l_2,m_2}\sum_{l,m}(-\mathrm{i}m)\tilde{q}^{l_2 m_2}_{lm} Q^{\mathrm{lab}}_{lm} \tilde{Q}^{\mathrm{co}}_{l_2 m_2} e^{-\mathrm{i}m\phi} \\ + \sum_{l_2 m_2}\sum_{l,m}(-\mathrm{i}m)\tilde{q}^{l_2 m_2}_{lm} Q^{\mathrm{lab}}_{lm} Q^{\mathrm{lab}}_{l_2 m_2} e^{-\mathrm{i}(m+m_2)\phi}\Bigg]\end{aligned} \tag{F1}$$

where

$$\tilde{q}^{l_2 m_2}_{lm} = \int \frac{\nabla'[\tilde{\mathcal{S}}_{lm}] \cdot \nabla'[\tilde{\mathcal{S}}_{l_2 m_2}]}{(2l+1)(2l_2+1)}\mathrm{d}V. \tag{F2}$$

The integrand involving the dot product of two gradients can be transformed using the following identity

$$\nabla u \cdot \nabla v = \frac{1}{2}[\nabla^2(uv) - v(\nabla^2 u) - u(\nabla^2 v)], \tag{F3}$$

where the Laplacian of $\mathcal{S}_{lm}$ is zero

$$\nabla^2 \mathcal{S}_{lm} = 0. \tag{F4}$$

Consequently, we can rewrite the expression as

$$\tilde{q}_{lm}^{l_2 m_2} = \int \frac{\nabla'^2[\mathcal{S}_{lm}\mathcal{S}_{l_2 m_2}]}{2(2l+1)(2l_2+1)} \mathrm{d}V. \tag{F5}$$

The product $\mathcal{S}_{lm} \cdot \mathcal{S}_{l_2 m_2}$ can be related to the product of two spherical harmonics $Y_{lm}(\vartheta,\varphi) \cdot Y_{l_2 m_2}(\vartheta,\varphi)$ as follows ([19], pp. 144)

$$Y_{lm}(\vartheta,\varphi) Y_{l_2 m_2}(\vartheta,\varphi) = \sum_L \frac{\sqrt{(2l+1)(2l_2+1)}}{\sqrt{4\pi(2L+1)}} \cdot C_{l0l_20}^{L0} C_{lml_2m_2}^{L,m+m_2} Y_{L,m+m_2}(\vartheta,\varphi), \tag{F6}$$

where $C_{lml_2m_2}^{LM}$ is the Clebsch-Gordan coefficient, and integer L ranges from $-|l-l_2|$ to $l+l_2$.

Therefore, Eq. (F5) can be further transformed as

$$\tilde{q}_{lm}^{l_2 m_2} = \sum_L \frac{C_{l0l_20}^{L0} C_{lml_2m_2}^{L,m+m_2}}{2\sqrt{4\pi(2l+1)(2l_2+1)(2L+1)}} \cdot \int \nabla'^2[r'^{l+l_2} Y_{L,m+m_2}^{\dagger}(\vartheta',\varphi')] \mathrm{d}V. \tag{F7}$$

Additionally, the integrand can be rewritten using the equation ([19], pp. 133)

$$\left[\nabla^2 - \frac{(n-l)(n+l+1)}{r^2}\right][r^n Y_{lm}] = 0 \tag{F8}$$

or

$$\nabla'^2[r'^{l+l_2} Y_{L,m+m_2}^{\dagger}(\vartheta,\varphi')] = (l+l_2-L)(l+l_2+L+1) \cdot r'^{l+l_2-2} Y_{L,m+m_2}^{\dagger}(\vartheta',\varphi'). \tag{F9}$$

Consequently,

$$\tilde{q}_{lm}^{l_2 m_2} = \sum_L \frac{(l+l_2-L)(l+l_2+L+1)}{2\sqrt{4\pi(2l+1)(2l_2+1)(2L+1)}} \cdot C_{l0l_20}^{L0} C_{lml_2m_2}^{L,m+m_2} \int r'^{l+l_2-2} Y_{L,m+m_2}^{\dagger}(\vartheta',\varphi') \mathrm{d}V. \tag{F10}$$

The leading $1\omega_s$ term in the torque, Eq. (F1), comprises the summation of the terms with $l=1, m=\pm 1$ and $l_2=1$ for the first term in the square brackets, and $l=1, m=\pm 1$ and $l_2=1$, $m_2=0$ for the second term. Considering these $\tilde{q}_{lm}^{l_2 m_2}$ components,

$$\tilde{q}_{11}^{10} = \tilde{q}_{1,-1}^{10} = 0, \qquad \tilde{q}_{11}^{11} = \tilde{q}_{1,-1}^{1,-1} = 0, \tag{F11}$$

and

$$\tilde{q}_{11}^{1,-1} = -(\pi/12)V, \qquad \tilde{q}_{1,-1}^{11} = -(\pi/12)V, \tag{F12}$$

the leading $1\omega_s$ torque can be expressed as

$$\tau_{1\omega_s} = \frac{\mu_0 \chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} (\mathrm{i}\tilde{q}_{1,-1}^{11} Q_{1,-1}^{\mathrm{lab}} \tilde{Q}_{11}^{\mathrm{co}} e^{\mathrm{i}\phi} - \mathrm{i}\tilde{q}_{11}^{1,-1} Q_{11}^{\mathrm{lab}} \tilde{Q}_{1,-1}^{\mathrm{co}} e^{-\mathrm{i}\phi}) \tag{F13}$$

The leading $2\omega_s$ term in the torque, Eq. (F1), is the summation of the terms with $m=\pm 2$, $l=2$, $l_2=1$ for the first term in the square brackets, and $m=m_2=\pm 1$, $l=l_2=1$ for the second term. Considering these components,

$$\tilde{q}_{22}^{11} = \tilde{q}_{2,-2}^{1,-1} = 0, \tag{F14}$$

$$\tilde{q}_{22}^{10} = \tilde{q}_{2,-2}^{10} = 0, \tag{F15}$$

and

$$\tilde{q}_{22}^{1,-1} = \frac{\sqrt{5}}{20\pi} \int (x' - \mathrm{i}y') \mathrm{d}V, \tag{F16}$$

$$\tilde{q}_{2,-2}^{11} = -\frac{\sqrt{5}}{20\pi} \int (x' + \mathrm{i}y') \mathrm{d}V, \tag{F17}$$

the leading $2\omega_s$ torque can be calculated as

$$\tau_{2\omega_s} = \frac{2\mu_0 \chi_{\mathrm{m}}}{1+\chi_{\mathrm{m}}} (-\mathrm{i}\tilde{q}_{22}^{1,-1} Q_{22}^{\mathrm{lab}} \tilde{Q}_{1,-1}^{\mathrm{co}} e^{-\mathrm{i}2\phi} + \mathrm{i}\tilde{q}_{2,-2}^{11} Q_{2,-2}^{\mathrm{lab}} \tilde{Q}_{11}^{\mathrm{co}} e^{\mathrm{i}2\phi}). \tag{F18}$$

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